

RAJKUMAR VIJAYAN

MSc Software Development Candidate | Technology | Data & Analytics | AI

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LinkedIn | GitHub

Professional Summary

MSc Software Development student at the University of Limerick with an academic foundation spanning **Data Science, Business Analytics and Software Development**, complemented by experience in **machine-learning data operations and business consulting**. Experienced in Python, SQL, Power BI, C++ and Java, with projects across LLM-based software testing, analytics and systems development. Interested in graduate opportunities across **Technology, Data & Analytics, AI and Product**, combining technical problem-solving with business understanding.

Education

University of Limerick

Limerick, Ireland

MSc Software Development (International Systems)

2025 – 2027

- Relevant study includes Software Testing & Inspection, Text Analytics & Natural Language Processing, Philosophy of Research, Data Structures & Algorithms and Software Development.
- **Dissertation:** An Empirical Evaluation of Large Language Models for Automated Software Test Generation Using Mutation Testing.

Thiagarajar School of Management

Madurai, India

PGDM – Business Analytics

2023 – 2025

- Developed practical knowledge of business analytics, data-driven decision-making, requirements analysis, stakeholder management and business problem solving.
- Applied business analytics concepts to data-driven problem solving, stakeholder-focused analysis and team-based projects.

B.S. Abdur Rahman Crescent Institute of Science and Technology

Chennai, India

Bachelor of Computer Applications (BCA) – Data Science

2017 – 2020

- Developed foundational knowledge in programming, databases, statistics, data analysis and data science concepts.

Professional Experience

Amazon Development Centre

Chennai, India

ML Data Associate

Feb 2022 – Jan 2023

- Analysed and curated more than **100,000 data records** used in machine-learning workflows while working against defined quality and accuracy standards.
- Performed data validation and quality checks to identify inconsistencies and improve the reliability of datasets used in downstream processes.
- Investigated data issues using structured analytical approaches, supporting consistent data quality while meeting productivity and accuracy targets.
- Collaborated within a data operations team and contributed to process improvements associated with a **30% increase in team productivity**.

VRBB Associates

India

Business Consulting Intern

May 2024 – Jul 2024

- Supported **requirements gathering and analysis** of client business needs across consulting assignments.
- Organised findings and documentation to translate business requirements into clear information for project discussions.
- Worked collaboratively to analyse business problems and support the development and communication of practical recommendations.

Technical & Analytics Projects

TestForge – LLM-Based Automated Software Test Generation | *Python / LLMs / Mutation Testing* **2026**
| **GitHub**

- Developing a dissertation-aligned framework to evaluate **Large Language Models for automated software test generation**.
- Uses **mutation testing** to evaluate whether generated tests detect injected software faults rather than relying only on test execution or code coverage.
- Designed to support empirical comparison of generated tests using measurable software-testing effectiveness criteria.

CRM Sales Pipeline Analysis | *Power BI / Business Analytics* | **GitHub**

2026

- Developed an interactive Power BI dashboard to analyse sales pipeline performance, customer segments and conversion patterns.
- Applied **RFM segmentation and funnel analysis** to investigate customer behaviour and identify commercial opportunities from sales data.
- Designed KPI-driven visualisations to translate raw business data into clear, decision-oriented insights for business users.

InferX – C++ Inference Scheduler | *C++20 / CMake / GoogleTest* | **GitHub**

2026

- Developed a C++20 inference scheduler implementing **dynamic batching, priority scheduling and thread-pool execution**.
- Implemented latency measurement using **p50, p95 and p99 metrics** to analyse scheduler performance and runtime behaviour.
- Structured the project using CMake and GoogleTest to support maintainability, automated testing and reliable development workflows.

Technical Skills

Programming: Python, Java, C++, JavaScript, SQL

Data & Analytics: Power BI, Pandas, NumPy, Scikit-learn, Microsoft Excel, Data Cleaning, Exploratory Data Analysis, RFM Segmentation, Cohort Analysis, Data Visualisation, Statistical Analysis, Machine Learning Fundamentals

Software Engineering: Object-Oriented Programming, Software Testing, Mutation Testing, Data Structures & Algorithms, Git, GitHub, CMake, GoogleTest

Tools & Platforms: Docker, Google Cloud, Streamlit, WSL

Business Analysis: Requirements Gathering, Business Analysis, Documentation, Stakeholder Communication, Client Needs Assessment, Problem Solving

Leadership & Extracurricular Activities

Entrepreneurship Cell – Thiagarajar School of Management

Madurai, India

Core Committee Member – IT & Promotions

2023 – 2025

- Helped organise entrepreneurship and business events involving more than **1,000 students**.
- Negotiated and coordinated approximately **15–20 sponsorships** while working with external stakeholders and cross-functional student teams.
- Supported venture events, case competitions and founder sessions, demonstrating communication, collaboration and organisational skills.

Professional Development

McKinsey Forward Program – Completed

- Completed professional development training focused on structured problem-solving, communication, adaptability and workplace effectiveness.

Work Eligibility

Stamp 2 student permission | Eligible to apply for **Stamp 1G graduate permission** upon successful completion of MSc (2027).